Chapter III.8

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8.1. Let $f : X \to Y$ be a continuous map of topological spaces. Let $\mathscr{F}$ be a sheaf of abelian groups on X, and assume that $R^i f_*(\mathscr{F}) = 0$ for all $i > 0$. Show that there is a natural isomorphism

$$H^\bullet(X, \mathscr{F}) \cong H^\bullet(Y, f_*\mathscr{F}).$$

(This is a degenerate case of the Leray spectral sequence.)

Proof. Note that we may regard X as a ringed space by setting $\mathscr{O}_X = \underline{\mathbb{Z}}$, so that $\mathfrak{Ab}(X) \cong \mathfrak{Mod}(X)$. We also have

$$H^0(X, \mathscr{F}) = \Gamma(X, \mathscr{F}) = \Gamma(Y, f_*\mathscr{F}) = H^0(Y, f_*\mathscr{F}).$$

Thus, it suffices to show that $H^\bullet(Y, f_*(-))$ is a universal δ-functor. f_* is exact by assumption, so it suffices by (1.3A) to show that $H^i(Y, f_*(-))$ is effaceable for each $i > 0$. Each injective object $\mathscr{I}$ of $\mathfrak{Ab}(X)$ is flasque by (2.4), so each $f_*\mathscr{I}$ is flasque by (II, Ex. 1.16d). Then $H^i(f_*\mathscr{I}) = 0$ for all $i > 0$ by (2.5), so we're done. □

8.2. Let $f : X \to Y$ be an affine morphism of schemes with X noetherian, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Show that the hypotheses of (Ex. 8.1) are satisfied, and hence that $H^\bullet(X, \mathscr{F}) \cong H^\bullet(Y, f_*\mathscr{F})$. (This gives another proof of (Ex. 4.1).)

Proof. It suffices to show that $R^i f_*(\mathscr{F}) = 0$ for all $i > 0$. Take $U \subset Y$ an affine open, so that $V := f^{-1}(U)$ is affine. (8.2) and (8.5) imply

$$R^\bullet f_*(\mathscr{F})|_U \cong R^\bullet(f|_V)_*(\mathscr{F}|_V) \cong H^\bullet(V, \mathscr{F})^\sim,$$

whence

$$R^i f_*(\mathscr{F})|_U \cong H^i(V, \mathscr{F})^\sim = 0$$

for all $i > 0$ by (3.5). It follows that $R^i f_*(\mathscr{F}) = 0$ for all $i > 0$, as desired. □

8.3. Let $f : X \to Y$ be a morphism of ringed spaces, let $\mathscr{F}$ be an $\mathscr{O}_X$-module, and let $\mathscr{E}$ be a locally free $\mathscr{O}_Y$-module of finite rank. Prove the *projection formula* (cf. (II, Ex. 5.1))

$$R^\bullet f_*(\mathscr{F} \otimes f^*\mathscr{E}) \cong R^\bullet f_*(\mathscr{F}) \otimes \mathscr{E}.$$

Proof. For $i = 0$, this is precisely the projection formula (II, Ex. 5.1d). f^* is right adjoint and thus commutes with inverse limits, and in particular finite direct sums. $f^*(\mathscr{O}_X) = \mathscr{O}_Y$, so this gives us $f^*\mathscr{E}$ locally free of finite rank. Then both $R^\bullet f_*(- \otimes f^*\mathscr{E})$ and $R^\bullet f_*(-) \otimes \mathscr{E}$ are δ-functors, so it suffices to show each is effaceable.

Clearly the latter is effaceable. Now, take and embedding $\mathscr{F} \hookrightarrow \mathscr{I}$ of $\mathscr{F}$ into an injective object of $\mathfrak{Mod}(X)$. (II, Ex. 5.1abc) gives us an isomorphism

$$\mathrm{Hom}(-, \mathscr{I} \otimes f^*\mathscr{E}) \cong \mathrm{Hom}(- \otimes (f^*\mathscr{E})^\vee, \mathscr{I}),$$

where the latter functor is exact since $(f^*\mathscr{E})^\vee$ is locally free of finite rank and $\mathscr{I}$ is injective. Then $\mathscr{I} \otimes f^*\mathscr{E}$ is injective, whence $R^\bullet f_*(- \otimes f^*\mathscr{E})$ is effaceable. The result follows from (1.3A). □